HUGO GABRIEL PEREIRA DE ALMEIDA

AI Engineering Student · Machine Learning & Quantitative Finance

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AI Engineering student at ESEO (2027), majoring in Machine Learning and Quantitative Finance. Proven track record building production-grade AI systems: deployed obstacle detection and map-based localization models at SNCF (work-study) and independently developed a reinforcement learning portfolio optimizer achieving a Sharpe ratio of 1.71. Currently engineering a low-latency HFT order-book in C++ and co-authoring a research paper on ML applied to nephrology. Seeking a 3-month international internship (from June 2026) to apply AI and quantitative skills in a high-performance engineering environment.

PROFESSIONAL EXPERIENCE

SNCF · Le Mans, France

Work-Study AI Engineer — AI Department *Sept 2025 – Present*

- Benchmarked and deployed multiple deep learning and computer vision models to achieve robust map-based localization and alignment for railway infrastructure.
- Engineered an obstacle detection pipeline combining odometry and CV models, improving detection reliability across diverse real-world operational conditions.
- Managed the full ML lifecycle: data preprocessing, model training, evaluation, iterative refinement, and production deployment.

NextWize · Remote

Portfolio Trading Analyst *Feb 2026 – Present*

- Actively managing and rebalancing a Euronext equity portfolio within the Trading Game, applying quantitative reallocation strategies to optimize risk-adjusted returns.

LEAD · Dijon, France

IT Intern — AI & Healthcare *Jun 2024 – Jul 2024*

- Contributed to a multidisciplinary AI & healthcare research project built, evaluated, and iterated on ML models with a focus on classification performance metrics.
- Developed a Python GUI to streamline data processing and model experimentation workflows for the research team.
- Work contributed to an ongoing research paper on ML applied to nephrology (publication forthcoming).

CHU — University Hospital · Dijon, France

IT Intern *Jun 2023 – Jul 2023*

- Developed a healthcare-themed educational game (Scratch), presented at DataKids, a multi-school outreach event for primary to high-school students — while co-organizing the event.

SELECTED PROJECTS

OptiFolio | RL-Driven Portfolio Optimizer *Oct 2025 – Feb 2026*

Tech: Python · PPO (Stable-Baselines3) · Gymnasium · NumPy · Pandas

- Designed and trained a PPO-based RL agent for dynamic asset allocation across GLD, USO, and URTH, achieved a Sharpe ratio of 1.71 vs. a buy-and-hold baseline.
- Implemented full reward-shaping and risk-control pipeline, benchmarked against classical strategies. Full technical report available on GitHub.

Quant-Strat | Quantitative Trading Strategy Library *Mar 2026 – Present*

Tech: Python · XGBoost · Scikit-learn · Pandas · Jupyter · Deep Learning (option pricing)

- Building a modular open-source library of independent algorithmic trading strategies, each following a structured research pipeline: hypothesis → data → feature engineering → model → backtest → performance analysis (Alpha, Beta, Sharpe, Volatility, Drawdown).
- First strategy: XGBoost return prediction using lagged returns, rolling statistics, and technical indicators, full backtest with performance evaluation.
- Additional modules developed: deep learning-based option pricing and statistical arbitrage strategies.

UDS-OB-HFT | High-Frequency Trading Engine *Feb 2026 – Present*

Tech: C++ · Unix Domain Sockets · Low-latency systems

- Engineering a low-latency order book and HFT client in C++ using Unix Domain Sockets for inter-process communication and latency simulation.
- Focus on memory efficiency, lock-free data structures, and microsecond-level execution performance.

Cerberus | AutoEncoder Fraud Detection *2025*

Tech: Python · PyTorch · Kaggle Credit Card Dataset

- Trained an Autoencoder on non-fraudulent transactions, detected fraud via reconstruction error (MSE) with automatic F1-optimized threshold selection.
- Achieved strong classification performance on a highly imbalanced dataset, evaluated with confusion matrix and full classification report.

TECHNICAL SKILLS

Languages:	Python · C · C++ · MATLAB · SQL · Git
AI / ML:	Reinforcement Learning (PPO) · Deep Learning (AutoEncoder, option pricing) · Computer Vision · Odometry · ML (regression, classification, gradient boosting, etc.)
Libraries:	PyTorch · TensorFlow · Stable-Baselines3 · XGBoost · Scikit-learn · NumPy · Pandas · Matplotlib · Gymnasium
Finance / Quant:	Portfolio optimization · Back testing · Risk metrics (Sharpe, Alpha, Beta, Volatility, Drawdown) · Algorithmic trading · Option pricing · Factor modelling
Systems:	Unix Domain Sockets · Low-latency C++ systems · STM32 embedded (C) · Real-time constraints · Memory management
Other:	Technical writing · Project management (MOOC) · Cross-functional communication · Multicultural environments

RESEARCH & PUBLICATION

Machine Learning Applied to Nephrology

Co-author · Research paper in preparation — expected end of May 2026

- Applying supervised machine learning models to clinical nephrology data to improve diagnostic support. Research builds directly on ML experiments and data pipelines developed during the LEAD internship (2024).
- Responsibilities include model selection, feature engineering on medical datasets, performance evaluation. The paper is finalized and currently being prepared for publication.

EDUCATION

ESEO · Angers, France

Engineering Degree — Information Systems & Artificial Intelligence *2022 – 2027*

- 5-year accredited engineering program (Bac+5), majoring in Artificial Intelligence and Information Systems.
- Key coursework: machine learning, deep learning, reinforcement learning, signal processing, software engineering, embedded systems (STM32), and systems programming.
- Academic projects spanning embedded C development, AI model deployment. Complemented by a 2-year work-study at SNCF.

Eifel · Dijon, France

High School Diploma — English European Section, Honours *2019 – 2022*

- Graduated with Honours in the English European stream, with advanced coursework in mathematics and sciences.

CERTIFICATIONS & LANGUAGES

Certifications:	TOEIC 925/990 (Mar 2025) · Nvidia: Getting Started with Deep Learning (Oct 2025) · Cambridge CAE (May 2022) · Yale: Financial Markets (in progress)
Languages:	Portuguese (Native) · French (Bilingual) · English (C1+, TOEIC 925) · Spanish (B2+)

INTERESTS & HOBBIES

Fine Watchmaking · Fashion · Motorsport · Basketball · Hiking · Golf · Travelling